

AARSHIA GUPTA

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EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO

La Jolla, CA

Bachelor of Science in Data Science | Minor in Business Analytics | Major GPA: 3.957 | Provost Honors Expected Graduation, June 2027

Relevant Coursework: Probabilistic Modeling & Machine Learning, Statistical Inference, Data Structures & Algorithms, Object-Oriented Programming, Data Management, Linear Algebra, Algorithmic Efficiency Principles, Graph Theory, Recursive Algorithms, D3

WORK EXPERIENCE

DATA SCIENCE INTERN

Gurugram, IND

Ernst & Young (EY)

June 2025 - Aug 2025

- **Forecasted demand** for **200K+** FMCG products, as measured by a 12% drop in **MAPE/RMSE**, using **Random Forest**, **SARIMA** models in Python to predict sales across multiple granularities (product, category, and region).
- Improved pipeline usability, as measured by uninterrupted weekly outputs, by engineering **Python-SQL ETL** workflows that integrated seasonality & promotions data and supported forecasts across product, category, and region levels through **time-series EDA**.
- Increased **model robustness**, measured by 18% out-of-sample results, by tuning RF estimators, SARIMA parameters, rolling-window validation.
- Enabled executive decision-making, as measured by adoption in 3 strategy meetings, by presenting forecasts, error drivers, and business scenarios.

DATA CONSULTING ANALYST

La Jolla, CA

marimo | Consulting@DS3 | [GitHub](https://github.com)

Oct 2024 - June 2025

- Developed open-source notebooks, measured by integration into **15K+** starred GitHub-repo, by teaching Polars, Matplotlib, Plotly, scikit-learn.
- Expanded **accessibility** (400+ users onboarded) by teaching reproducible data pipelines & version control, onboarding to a Jupyter alternative.

DATA SCIENCE INTERN

New Delhi, IND

NRV DesignX

Jul 2024 - Sep 2024

- Improved chatbot pilot adoption, as measured by 25% more user engagement, by **researching** GenAI vs. traditional LLM trade-offs.
- Boosted predictive accuracy, as measured by 20% validation lift, via **preprocessing** pipelines (cleaning, normalization, transformation) for big datasets.
- Supported stakeholder decisions, measured by adoption in 2 meetings, by **training** regression, classification models evaluated with accuracy/F1-score.

RESEARCH ANALYST INTERN

New Delhi, IND

Osmosys Incorporation

May 2022 - Jul 2022

- Increased client satisfaction by 15%, as measured in surveys, by **analyzing** unstructured market data in Excel & delivering Tableau insight dashboards.

PROJECTS

STEAM GAME RECOMMENDER SYSTEM | Python, scikit-learn, PyTorch, SBERT, Logistic Regression

Dec 2025

- Built **hybrid recommender** framed as binary classification, combining statistical features, SBERT semantic embeddings, user one-hot encoding.
- Compared **7 models** (LogReg, CF, RF, neural nets), with best (**LogReg + SBERT + OHE**) achieving **AUC 0.9279**, a ~20% lift over baseline.
- Showed semantic embeddings solve cold-start for new games while OHE provides strong user-level personalization.

MEAL TIME PREDICTOR | Python, scikit-learn, Logistic Regression, GridSearchCV

Feb 2025 - Mar 2025

- Built a recipe-duration **prediction model** by processing **83K+ recipes** with structured metadata (ingredients, macronutrients, procedural steps).
- Doubled recall by EDA, engineering nutritional/procedural features, applying normalization (Quantile Transformation, Scaling, Binarization).
- Improved performance (**F1: 0.235 → 0.531**) by training **logistic regression with 5-fold GridSearchCV**, revealing meal complexity indicators.

WHAT'S YOUR SLEEP STORY? | Javascript (D3), Euclidean Distance, UX Engineering

May 2025 - Jun 2025

- Designed, implemented **web application & visualization dashboard** for 180×20×9 biometric dataset (200+ raw CSVs) with D3, HTML/CSS.
- Improved interpretability of multidimensional sleep data, **measured by 20+ tester engagement**, through data ingestion pipeline & modular UI.
- Delivered **real-time personalized insights** by clustering user similarity (Euclidean distance) & interactive “Fix Sleep” sliders with UX animation.

SENTIMENT ANALYSIS ON IRA TWEETS : NLP PROJECT | Python, NLTK, Vader, BeautifulSoup, Regex, NLP

Jan 2024

- Analyzed 90K+ IRA tweets by building **NLP pipeline**, automating scraping (BeautifulSoup), cleaning (Regex), bot detection, ensuring dataset quality.
- Identified sentiment and emotion trends by applying **tokenization** & Vader scoring, presenting insights through keyword clouds & emotion clusters.

LEADERSHIP

SOCIAL & FUNDRAISING DIRECTOR (Promoted Jan 2025)

La Jolla, CA

Data Science Student Society @ UC San Diego

Oct 2024 - present

- Expanded engagement across 500+ member org., as measured by record sponsorships & partnerships, by leading outreach & largest fundraising goals.
- Promoted data science **accessibility**, measured by 150+ participants, delivering ML workshops & organizing DataHacks hackathon (~800 attendees).

DIRECTOR & DESIGN HEAD

New Delhi, IND

Exun, Technology Club @ Delhi Public School, R.K. Puram

Jun 2021 - Apr 2023

- Expanded Exun symposium to **15K+ participants** from 45+ schools by organizing 20+ events & launching Girls in Tech initiatives, STEM series.
- Boosted visibility of flagship CTF to **2K+ users** (15 countries), measured by engagement, by leading 15 members to design, implement UI, workflows.

SKILLS

Programming: Python (NumPy, pandas, matplotlib, scikit-learn, Polars, Dask), Java, C++, JavaScript (HTML/CSS, D3), SQL (PostgreSQL), MATLAB, Power BI

SWE: Data Structures & Algorithms, Object-Oriented Programming, Software Development Lifecycle, Testing & Debugging, Version Control (Git), Unix/Linux

Development & Systems: Web Application Development, APIs, Data Pipelines, Distributed Systems, AWS (S3, EC2), Apache Spark, Performance Optimization

Applied Areas & Other: NLP, Recommender Systems, Data Viz, Quantitative Analysis, Generative AI, Predictive Modeling, Business Insights, Data Modeling